Chapter 2

Oscillations

Oscillatory Motion

Periodic motion: *is motion of an object that regularly returns to a given position after a fixed time interval.*

Several types of periodic motion in everyday life:

- Earth's rotation, swinging pendulum of a clock, a person bouncing on a trampoline, a vibrating guitar string, and a mass on a spring.

a special kind of periodic motion occurs in mechanical systems when the force acting on an object is proportional to the position of the object relative to some equilibrium position.

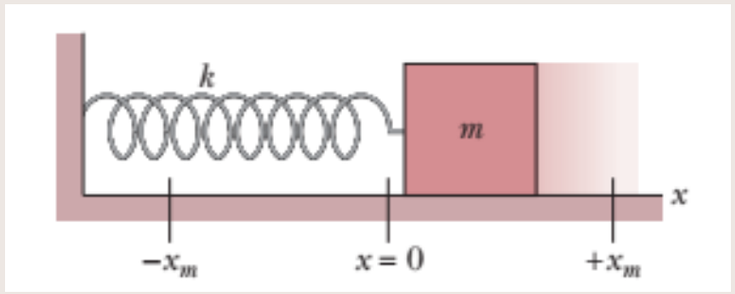
If this force is always directed toward the equilibrium position, the motion is called ***simple harmonic motion***.

Simple Harmonic Motion

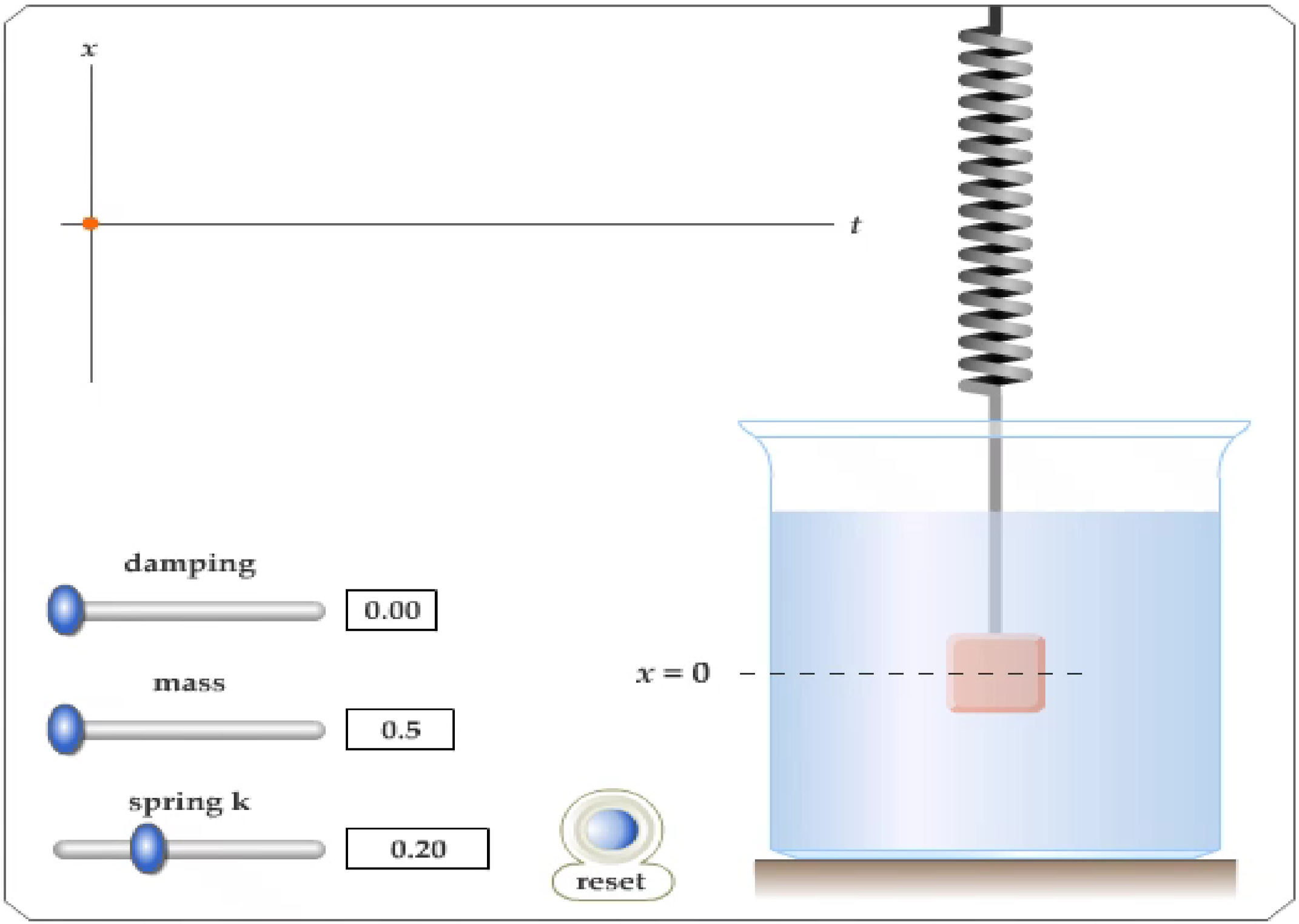
If a force acting on a body is proportional to the displacement and this motion repeating itself back and forth from its equilibrium position, this motion is named harmonic motion or periodic motion.

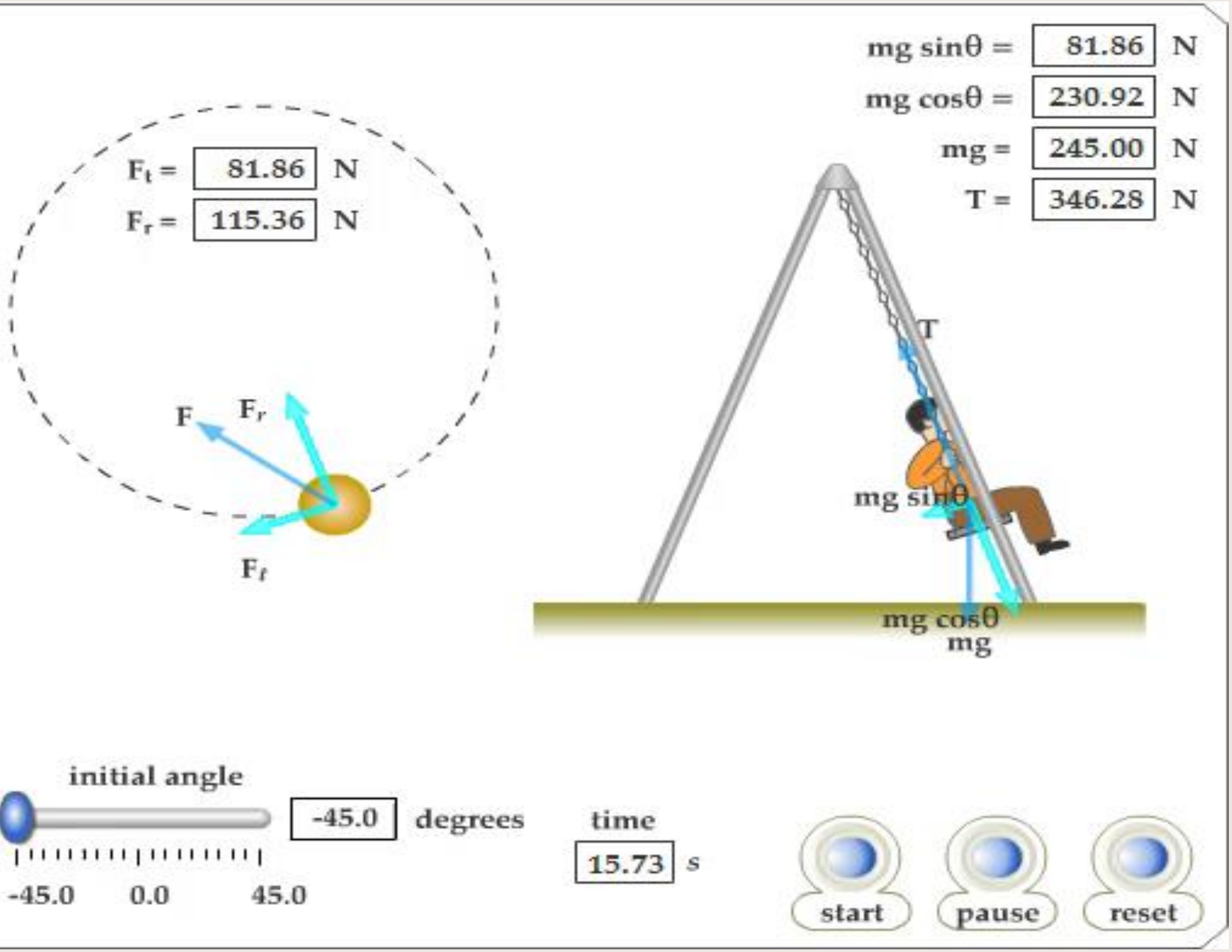
Periodic Motion and Oscillation

Some examples of periodic and oscillation motion:



1. **block attached to a spring**
2. **motion of a pendulum**
3. **vibrations of a stringed musical instrument**
4. **Oscillation of molecules in a solid**


x
t
damping
0.00
mass
0.5
spring k
0.20
reset
x = 0

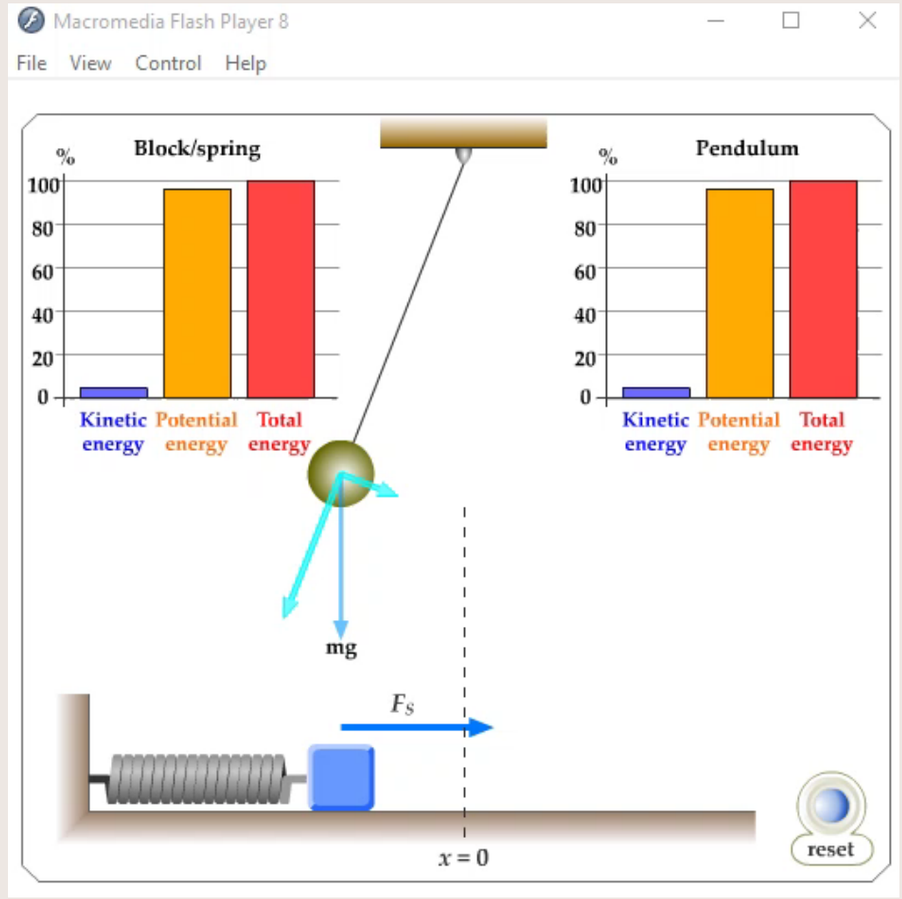

mg sinθ = 81.86 N
mg cosθ = 230.92 N
mg = 245.00 N
T = 346.28 N
F_t = 81.86 N
F_r = 115.36 N
F
F_r
F_t
T
mg sinθ
mg cosθ
mg
initial angle
-45.0 degrees
-45.0
0.0
45.0
time
15.73 s
start
pause
reset

Simple Harmonic Motion

- The force of the motion is directed toward the equilibrium position

- This motion repeat itself at regular intervals of time (Periodic Time)

- The acceleration of this motion is proportional to its displacement from its equilibrium position.

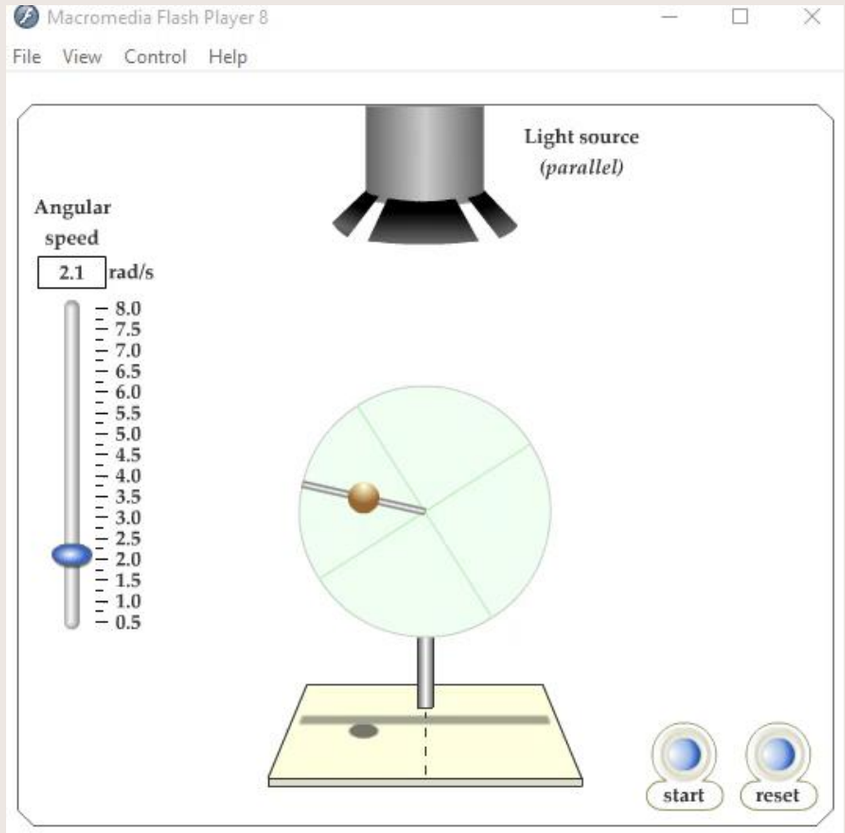


$$acceleration \propto displacement$$

Equation of oscillation motion

A particle moving along the x-axis exhibits simple harmonic motion when *X*, the particle's displacement from equilibrium, varies in time according to the relationship.

$$X = A\cos(\omega t + \varphi)$$



***A* is called amplitude**

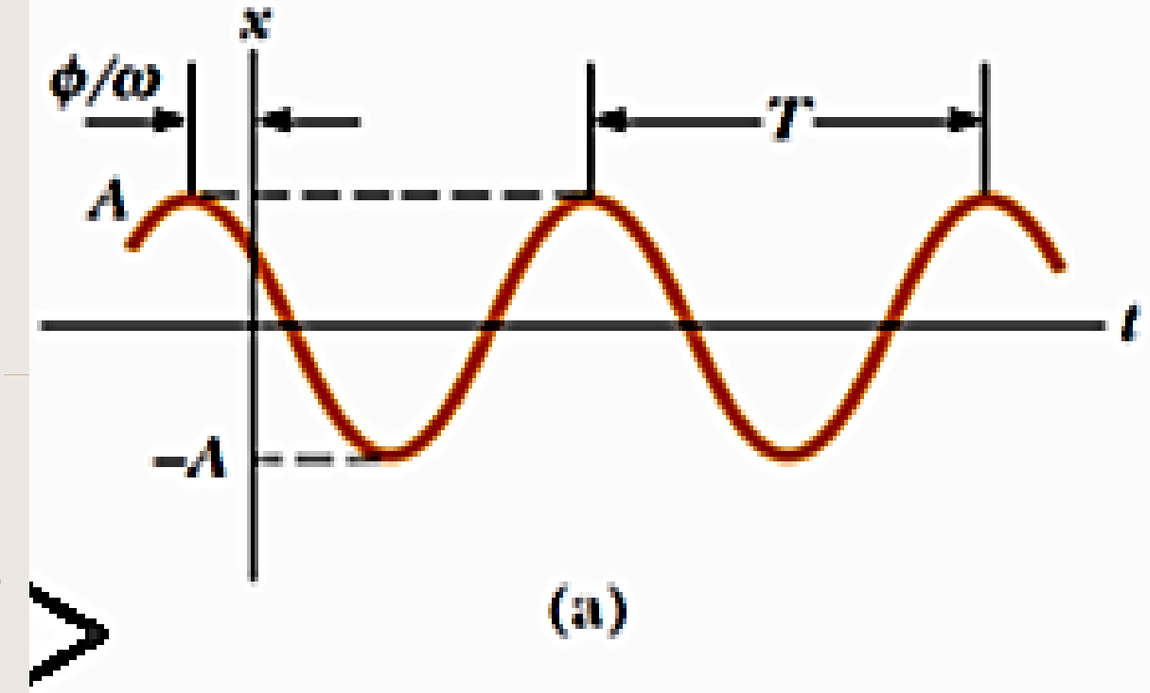
It is the maximum displacement of the particle in either the +ve or -ve X −direction

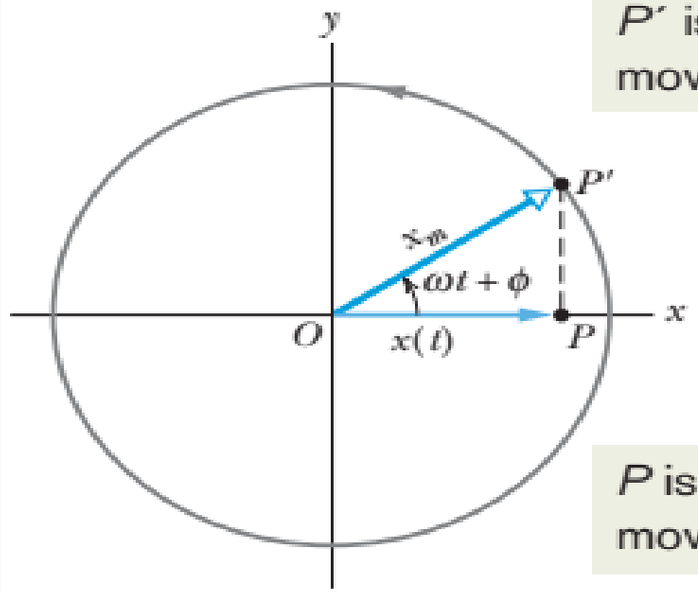
ω is called the angular frequency (rad/s)

- **It is number of cycles in one second particle**

φ is called the phase constant (phase angle)

- **It is determined by the initial displacement and velocity of the particle.**
- **The quantity $(\omega t + \varphi)$ is called the phase of the motion and is useful in comparing the motions of two oscillators.**





P′ is a particle moving in a circle.

P is a projection moving in SHM.

The oscillation equation is a periodic trigonometric function and repeating itself every complete cycle 2π which equivalent to the time of complete cycle (ωT), where T is perodic time

$$(\omega t + \varphi) + 2\pi = \omega(t + T) + \varphi$$
$$\omega t + \varphi + 2\pi = \omega t + \omega T + \varphi$$

$$\therefore \omega = 2\pi/T \qquad or \qquad \therefore \omega = 2\pi f$$

- **The inverse of the period is called the frequency f of the motion**

$$\therefore f = 1/T = \frac{\omega}{2\pi}$$

Frequency (f)

It is number of complete cycles per unit of time. Its unit is (Hz=1/s)

Peroidic Time

It is time needed to make one cycle. Its unit is second (s).

Velocity and acceleration of simple harmonic motion

The oscillating velocity

- **To obtain the oscillation velocity of a particle, So you can differentiate the displacement equation with respect to time.**

$$X(t) = A\cos(\omega t + \varphi)$$

$$v = \frac{dX}{dt} = \frac{d}{dt}(A\cos(\omega t + \varphi))$$

$$\therefore v = -A\,\omega\,\sin(\omega t + \varphi) \qquad \therefore v_{max} = -A\omega$$

The acceleration of the particle is

- **To obtain the acceleration of oscillating particle, So you can differentiate the velocity equation with respect to time.**

$$a = \frac{dv}{dt} = \frac{d}{dt}(-A\,\omega\,\sin(\omega t + \varphi))$$

$$\therefore \mathrm{a} = -A\,\omega^2\cos(\omega t + \varphi) \qquad \therefore \mathrm{a} = -\omega^2(A\cos(\omega t + \varphi))$$

$$a_{max} = -A\omega^2 \qquad a = -\omega^2 X$$

The force of SHM

$$\textbf{Force} = \textbf{Mass of particle} \times \textbf{acceleration}$$

$$\textbf{Force of SHM} = \textbf{Mass of particle} \times \textbf{acceleration of SHM}$$

$$\because a_{SHM} = -\omega^2 X$$

$$\because F = m\, a_{SHM} = -\,\mathrm{m}\,\omega^2 X$$

$$\because F = -\mathrm{m}\ \omega^2 X$$

Example:

An object oscillates with simple harmonic motion along the x axis. Its displacement from the origin varies with time according to the equation:

$$X = (4.00\,m)\,cos(\pi t + \pi/4)$$

where t is in seconds and the angles in the parentheses are in radians.

(a) Determine the amplitude, frequency, and period of the motion.

(b) Determine the maximum speed and maximum acceleration of the object.

Solution:

(a)

$$A = 4.00\,m$$

$$\because \omega t = \pi t \quad \therefore \ \omega = \pi \quad \rightarrow \ \pi = 2\pi f \qquad \therefore f = 0.5\,Hz$$

$$T = \frac{1}{f} = \frac{1}{(0.5)} = 2\,sec$$

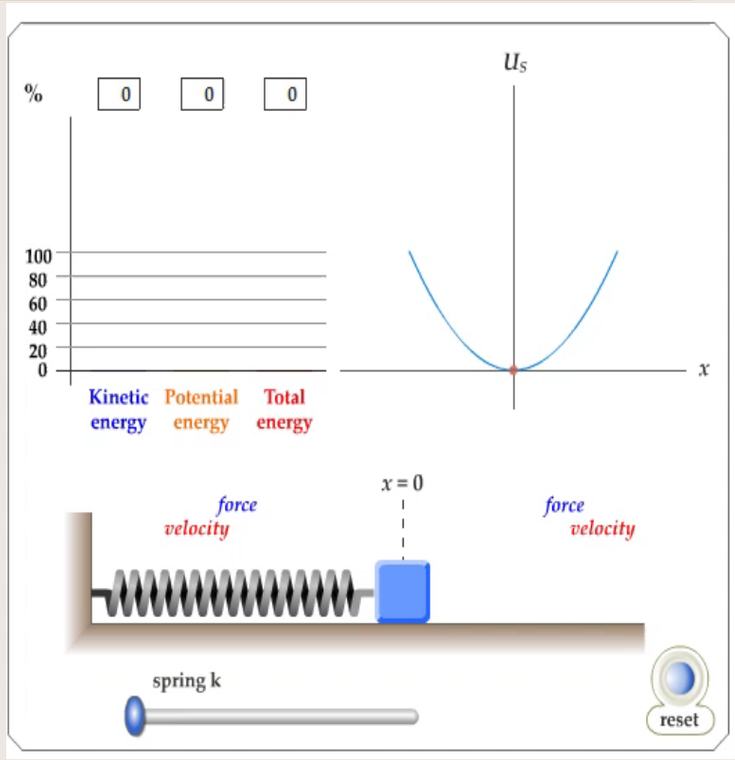
(b)

$$v_{max} = \omega A = (\pi)(4\,m) = 12.6\,ms^{-1}$$

$$a_{max} = \omega^2 A = (\pi)^2(4\,m) = 39.5\,ms^{-2}$$

Simple Harmonic Motion in The elastic spring

If block of mass *m* attached to the end of an elastic spring, during the spring is stretched or compressed as in graph.



when the block is displaced to a position *x*, the spring exerts on the block a force that is proportional to the position and given by Hooke's law:

$$F = -kx$$

where *k* is the spring constant which measured the stiffness of the spring. and *x* is the displacement of the object from its equilibrium position.

<u>The negative sign means that</u> *the force* exerted by the spring is always directed *opposite* the *displacement* of the object.

periodic time of oscillating object attached with elastic spring

Applying 2nd law of Newton $F = ma$

Also, the acceleration of SHM is $a = -\omega^2 x$

Then the force of SHM is

$$F = -m\,\omega^2 x \qquad (1)$$

Also, The restoring force of Hook's is $F = -kx$ (2)

then, equating the eq. (1) with eq. (2)

$$-k\,x = -m\,\omega^2 x$$

$$\therefore k = m\,\omega^2 \quad \rightarrow \quad \therefore \omega = \sqrt{\frac{k}{m}} \qquad \because \omega = 2\pi f = \frac{2\pi}{T}$$

$$\therefore T = 2\pi\sqrt{\frac{m}{k}} \qquad \textbf{or} \qquad \therefore f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

Example:

A car with a mass of 1300 kg is constructed so that its frame is supported by four springs. Each spring has a force constant of 20000 N/m. If two people riding in the car have a combined mass of 160 kg, find the frequency of vibration of the car after it is driven over a pothole in the road.

Solution:

The total mass = (1300+160) = 1460 kg

The mass act on one spring = 1460/4 = kg

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{(2000)}{\left(\frac{1300+160}{4}\right)}} = 1.18\ s^{-1}$$

Example:

A block with a mass of 200 g is connected to a light spring for which the force constant is 5 N/m and is free to oscillate on a horizontal, frictionless surface. The block is displaced 5 cm from equilibrium and released from rest (a) Find the period of its motion. (b) Determine the maximum speed of the block. (c) What is the maximum acceleration of the block?

Solution:

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{(200\times10^{-3}\ kg)}{(5\ Nm^{-1})}} = 1.26\ s$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{(1.26\ s)} = 5\ rad/s$$

$$v_{max} = \omega A = \left(5\frac{rad}{s}\right)(5\times10^{-2}\ m) = 0.25\ ms^{-1}$$

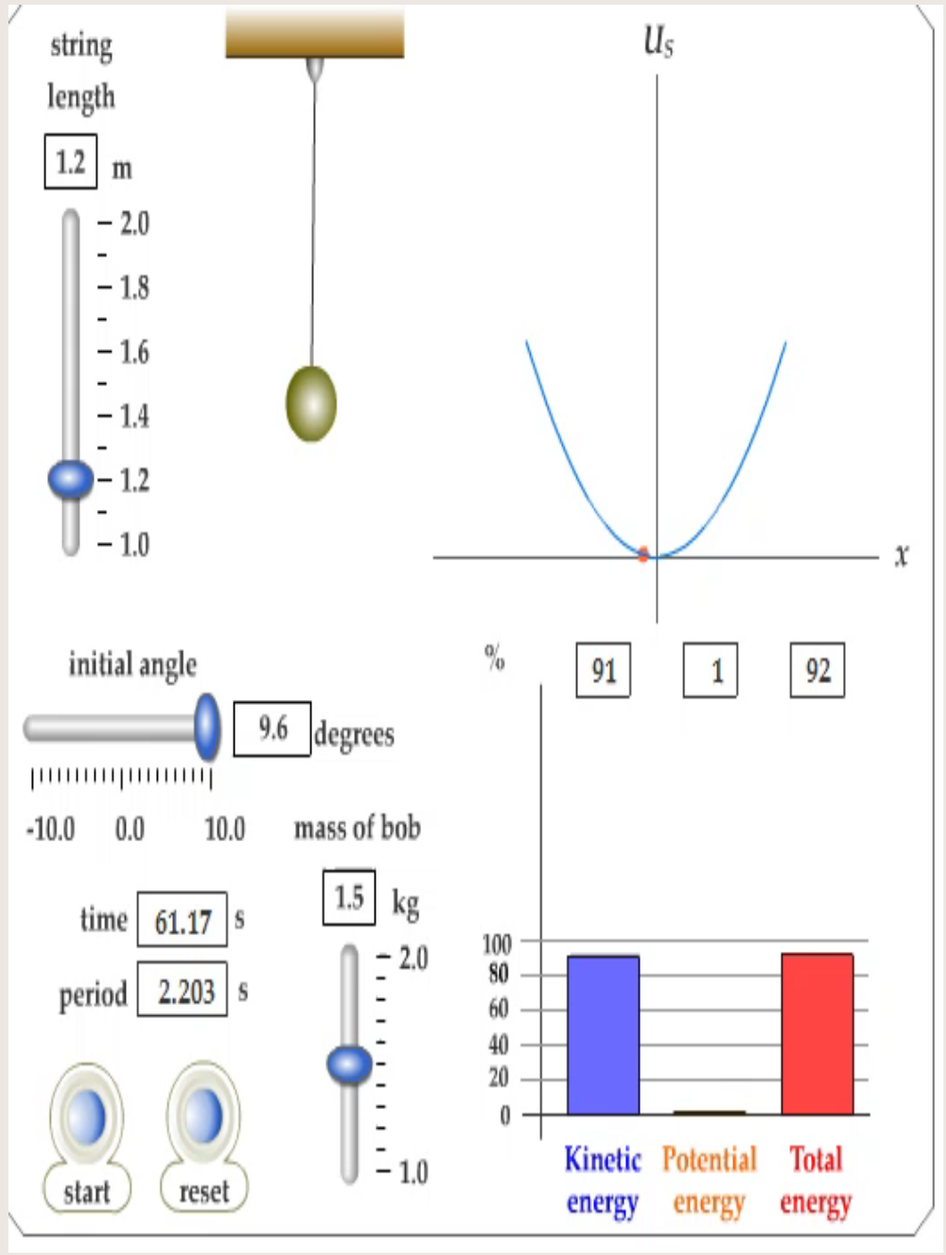
$$a_{max} = \omega^2 A = \left(5\frac{rad}{s}\right)^2(5\times10^{-2}\ m) = 1.25\ ms^{-2}$$

Simple Harmonic Motion in The Simple Pendulum

The simple pendulum is another mechanical system that exhibits periodic motion.

It consists of a particle-like bob of mass m suspended by a light string of length L that is fixed at the upper end as shown in Figure.

The motion occurs in the vertical plane and is driven by the gravitational force. We shall show that, provided the angle u is small (less than about 10º), the motion is very close to that of a simple harmonic oscillator.



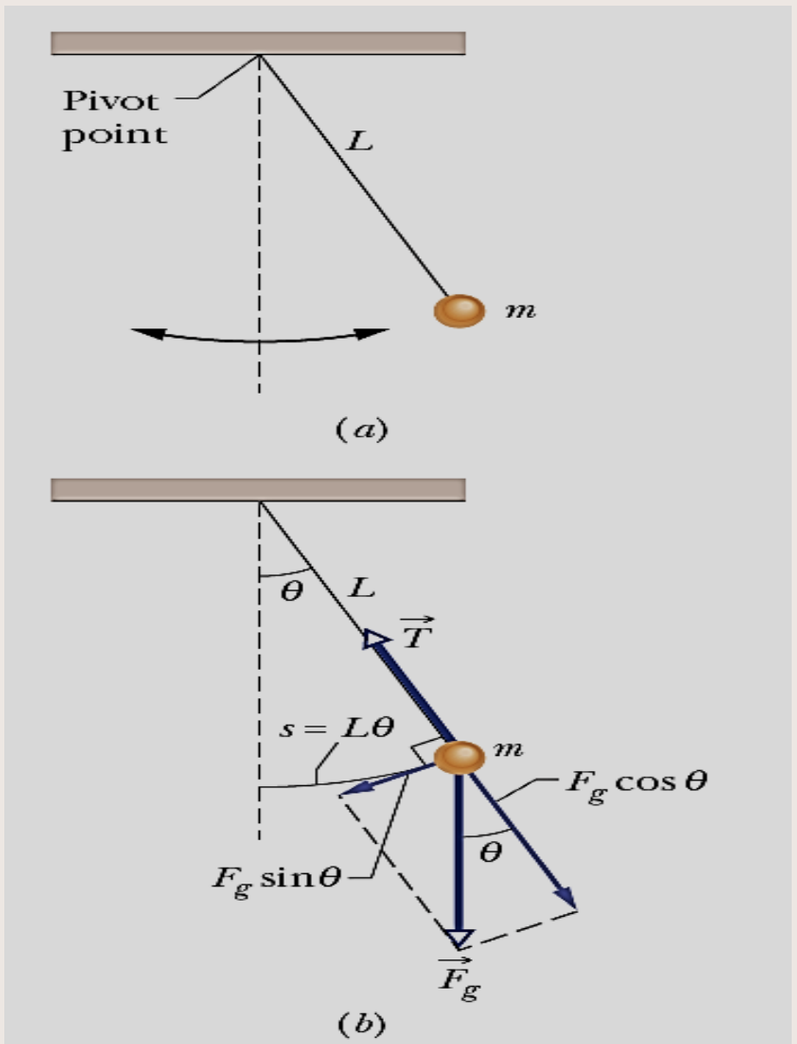
periodic time of simple pendulum

From the figure, after analysis the weight force of solid sphere

1- The restoring force is $F_t = -m\,g\,\sin\theta$

$$if\,\theta\ is\ very\ small \Rightarrow \sin\theta \approx \theta \approx \frac{X}{L}$$

$$F_t = -m\,g\,\frac{X}{L} \qquad (1)$$



2- The SHM force is

$$F_{SHM} = -m\,\omega^2\,X \qquad (2)$$

$$-m\omega^2 X = -m\,g\,\frac{X}{L} \qquad \therefore \omega = \sqrt{\frac{g}{L}}$$

$$\therefore \omega = 2\pi f = \frac{2\pi}{T} \qquad \therefore \frac{2\pi}{T} = \sqrt{\frac{g}{L}}$$

$$\therefore T = 2\pi\sqrt{\frac{L}{g}} \qquad \therefore f = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$$

Example:

Using a small pendulum of length 0.171 m, a geophysicist counts 72 complete swings in a time of 60 s. What is the value of *g* in this location?

Solution:

$$T = \frac{\text{total time}}{\text{total number of oscillations}} = \frac{(60\ \text{s})}{(72\ \text{osscilations})} = 0.833\ \text{s}$$

$$\mathrm{T} = 2\pi\sqrt{\frac{\mathrm{L}}{\mathrm{g}}} \Rightarrow \qquad \mathrm{g} = \frac{4\pi^2\mathrm{L}}{\mathrm{T}^2}$$

$$\mathbf{g} = \frac{\mathbf{4\pi^2(0.171\ m)}}{\mathbf{(0.833\ s)^2}} = 9.711\ ms^{-2}$$